

Counterexamples and Pitfalls: Slutsky's Theorem

10 exercises with hints only. The goal is to understand why the hypotheses of Slutsky's theorem are necessary.

Exercise 1. Give an example where $X_n \rightarrow X$ in probability, but Y_n does not converge in probability to any constant. Show that $X_n + Y_n$ need not converge to $X + c$.

Hint: Try $Y_n = (-1)^n$ as a deterministic sequence.

Exercise 2. Construct $X_n \rightarrow X$ in probability and $Y_n \rightarrow Y$ in probability where Y is non-constant. Determine the limit of $X_n + Y_n$ and explain why this is outside the usual statement of Slutsky.

Hint: Take $X_n = X$ and $Y_n = Y$ for all n .

Exercise 3. Find $X_n \rightarrow 0$ in probability and $Y_n \rightarrow 0$ in probability such that X_n/Y_n is not even defined with high probability.

Hint: Let Y_n equal 0 with positive probability.

Exercise 4. Show that the condition $c \neq 0$ in the division theorem cannot be removed.

Hint: Consider $X_n = 1$ and $Y_n = 1/n$.

Exercise 5. Give an example where $X_n \rightarrow X$ almost surely but Y_n oscillates and has no probability limit. Show that $X_n Y_n$ fails to converge.

Hint: Use a deterministic oscillating sequence for Y_n .

Exercise 6. Construct $X_n \rightarrow 0$ in probability and $Y_n \rightarrow \infty$ (deterministically). Investigate the behavior of $X_n Y_n$.

Hint: Choose $X_n = 1/n$ and $Y_n = n$, or modify to obtain different limits.

Exercise 7. Find sequences X_n and Y_n such that $X_n \rightarrow 0$ and $Y_n \rightarrow 0$ in probability but $X_n Y_n$ converges to a nonzero constant after suitable rescaling.

Hint: Think about $n X_n Y_n$.

Exercise 8. Give an example showing that convergence in distribution alone is insufficient for the division form of Slutsky.

Hint: Look for denominators approaching zero in distribution.

Exercise 9. Suppose $X_n \rightarrow X$ in probability and $Y_n \rightarrow c$ in distribution only. Determine whether Slutsky's conclusion must hold. Either prove it or find a counterexample.

Hint: Check carefully which mode of convergence is actually needed.

Exercise 10. Construct a sequence Y_n converging in probability to 0 for which $1/Y_n$ does not converge in probability to any finite random variable.

Hint: Take $Y_n = 1/n$.